

Manual Test Checklist

ECITY — Mobile Shop Management Web App

Version 1.1 · 2026

What a person still has to judge, module by module

Companion documents: PRD, Module Breakdown, Engineering Design, Deployment Guide, Database Guide

How to Use This

Automated tests cover the mechanics — 481 unit and integration tests, 912 browser tests across four screen sizes. This checklist covers what a person still has to judge: does it *feel* right, does the wording make sense, does the shop's actual workflow survive contact with the screen.

Work through a module's section after it is built, and again before go-live. Tick as you go. Anything that fails is a bug, not a note.

One section per module. M1–M14 append to this file as they ship.

Before you start

```
docker compose up -d      # or open Postgres.app
npm run db:migrate
npm run db:seed           # resets the three test accounts
npm run dev               # http://localhost:3000
```

To return to a known-clean state at any point, re-run `npm run db:seed`. It upserts, so it is safe to repeat.

Test accounts

Email	Password	Role	Branch scope
admin@ecity.local	ChangeMe!2026	Admin / Owner	All branches (14 permissions)
manager@ecity.local	ChangeMe!2026	Branch Manager	MAIN only (7 permissions)
staff@ecity.local	ChangeMe!2026	Staff	MAIN only (3 permissions)

ChangeMe!2026 is only the fallback the seed uses when `SEED_PASSWORD` is not set — which is the normal case on your own machine. **On staging or any deployed server the password is whatever `SEED_PASSWORD` was set to** in that server's `.env`; the accounts above are otherwise identical. Read it back with `grep SEED_PASSWORD ~/app/.env`.

Use a **private/incognito window** to hold a second session, so you can be two users at once without signing out.

These accounts must never exist in production. They are flagged `mustChangePassword` and are created only by the seed.

M0 — Foundations

Delivers. PRD FR-1.1 – FR-1.5, FR-31.1, plus branch context and the authorisation layer everything else depends on.

Signing in as the admin exercises perhaps a tenth of this module. The interesting part is what the *other two* users cannot do.

1. Authentication

Start signed out, at `/login`.

- [] Wrong password for a real account → *"Email or password is incorrect."*
- [] A made-up email with any password → **the identical message**. It must not reveal whether the account exists
- [] Empty email or password → inline validation, no request sent

- [] Visit `/settings/users` directly while signed out → redirected to `/login`
- [] Correct credentials → lands on `/dashboard`
- [] Avatar menu → **Sign out** → back at `/login`
- [] After signing out, the browser Back button does **not** restore the app
- [] Sign in again in a second browser → both sessions work independently

Password reset

Email is not wired up until M14, so the reset link is shown on screen in development.

- [] `/forgot-password` → enter `admin@ecity.local` → confirmation appears
- [] A **yellow development box** shows the reset link
- [] Enter an address that does not exist → **the same confirmation**, no hint that the account is unknown
- [] Follow the link, set a new password → redirected to `/login`
- [] The **old** password no longer works
- [] The **new** password works
- [] Open the same reset link a second time → refused as already used
- [] Reset the password back to `ChangeMe!2026`, or re-run `npm run db:seed`

Logout

- [] Get the password wrong 8 times in a row → *"Too many failed attempts."*
- [] Wait 15 minutes, or `npm run db:seed`, to clear it

2. Branch scoping

The heart of M0. Sign in as each user and open the branch switcher in the header.

- [] **admin** — sees *All branches (consolidated)*, MAIN and NORTH
- [] **manager** — sees **MAIN only**. No NORTH, and no "All branches"
- [] **staff** — sees MAIN only
- [] As admin, switch between branches → the dashboard's *Branch context* card updates (active branch, accessible count, consolidated yes/no)
- [] The chosen branch survives a page reload
- [] On a phone-width window the switcher shows the branch **code**, not the full name, and the header still fits

3. Permissions

Check what is **missing**, not what is present.

User	Sidebar should show
admin	Dashboard, Users, Roles, Audit log
manager	Dashboard, Users, Audit log — no Roles
staff	Dashboard only

- [] Each user's sidebar matches the table
- [] As **staff**, type `/settings/roles` into the address bar → redirected, not shown
- [] As **staff**, type `/settings/users` → redirected
- [] As **manager**, `/settings/roles` → redirected
- [] The dashboard's *Your permissions* card lists 14 / 7 / 3 entries respectively

4. Server-side authorisation

A hidden menu item is not a control. This step proves the server refuses the request, not just that the button was hidden.

Sign in as **staff** in the browser, then copy the session cookie from DevTools → Application → Cookies → `ecity_session`, and run:

```
COOKIE='ecity_session=PASTE_VALUE_HERE'
curl -s -o /dev/null -w '%{http_code}\n' localhost:3000/api/users -H "Cookie: $COOKIE"
curl -s -o /dev/null -w '%{http_code}\n' localhost:3000/api/audit -H "Cookie: $COOKIE"
curl -s -o /dev/null -w '%{http_code}\n' localhost:3000/api/roles -H "Cookie: $COOKIE"
```

- [] All three return **403**, not data
 - [] The body reads `{"error":"...", "code":"FORBIDDEN"}`
 - [] With no cookie at all → **401**
 - [] As **manager**, forcing another branch is refused:
- ```
curl -s -o /dev/null -w '%{http_code}\n' -X POST localhost:3000/api/session/branch \
-H "Cookie: $COOKIE" -H 'content-type: application/json' -d '{"branchId":2}'
```
- [ ] Returns **403** — branch NORTH is not theirs
  - [ ] `{"branchId":null}` (the consolidated view) also returns **403**

## 5. User management

As **admin**, Users → Add user.

- [ ] Password under 10 characters → rejected inline
- [ ] An email that already exists → *"A user with this email already exists."*
- [ ] No role chosen → *"Choose a role."*
- [ ] Create a Staff user assigned to **NORTH** only
- [ ] They appear in the user list with the right role and branch badge
- [ ] Sign in as the new user → branch switcher shows **NORTH only**
- [ ] Their first sign-in is flagged "must change password"

### Roles

As **admin**, Settings → Roles.

- [ ] Select **Staff**, tick *View the audit log*, Save
- [ ] Sign in as staff → **Audit log now appears** in their sidebar
- [ ] Any staff session open elsewhere was **signed out** when the role changed, rather than keeping stale permissions until next login

- ☐ Untick it and Save again
- ☐ Try to remove *Create and edit users* from **Admin** → refused, so the business cannot lock itself out
- ☐ System roles cannot be deleted

## 6. Audit log

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As **admin**, Settings → Audit log.

- ☐ Everything done above appears: `LOGIN`, `LOGIN_FAILED`, `LOGOUT`, `CREATE`, `PERMISSION_CHANGED`, `PASSWORD_RESET_COMPLETED`
- ☐ Each row shows who, what, when
- ☐ Expanding a "*n field(s)*" row shows old → new values
- ☐ A password hash **never** appears in a changes payload
- ☐ Paging works if there are more than 50 entries

The log is append-only, enforced by the database rather than by code:

```
psql ecity -c "update audit_log set summary='tampered' where id=1;"
psql ecity -c "delete from audit_log where id=1;"
```

- ☐ Both fail with `audit_log is append-only`

## 7. Responsive layout

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Resize the window, or use DevTools device mode. Check at ~375px, ~810px and full width.

- ☐ Below ~768px the sidebar is replaced by a **hamburger** top-left
- ☐ Tapping it opens a drawer; choosing a destination closes it
- ☐ On **Users**, the table becomes **one card per user**
- ☐ On **Audit log**, entries become cards
- ☐ Nothing scrolls sideways at any width
- ☐ Buttons and links are comfortably tappable with a thumb
- ☐ The **Roles** editor is usable — role list becomes a horizontal strip
- ☐ Forms stack, with full-width buttons

## 8. Theme

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- ☐ Primary colour is **black** throughout (buttons, active nav item)
- ☐ Set `APP_THEME=navy` in `.env`, restart → the whole app is blue, with no other visual change
- ☐ An invalid value such as `APP_THEME=nonsense` → falls back to black rather than breaking
- ☐ Set it back to `onyx`

## What M0 deliberately does not include

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Do not raise these as bugs:

- No products, inventory, IMEIs, customers or suppliers — those are M1 and M2
- The dashboard shows branch context and permissions, **not** sales figures. The real dashboard is M10
- No branch create/edit screen — M1. The two seeded branches are fixed
- Password reset emails print to screen instead of sending — M14
- Notifications, reports, exports — later modules

## M1 — Master Data

**Delivers.** PRD FR-2.1 – FR-2.4, FR-3.1, FR-3.2, FR-3.8, FR-5.10, FR-6.6, FR-10.1.

The shop gets configured and the people it trades with exist. Nothing here moves stock or money yet — these are the pickers and directories the rest of the system draws on.

**Customer** = someone who buys from the shop. **Supplier** = someone the shop buys from. Both are shared across every branch.

### 1. Business settings

Sign in as **admin**, go to **Settings** → **Business**.

- [ ] Four tabs: Profile, Tax, Payments, Expenses
- [ ] On a phone the tabs scroll sideways rather than wrapping

#### Profile

- [ ] Edit the business name, save, reload — the change persisted
- [ ] Enter a malformed GST number → *"Enter a valid 15-character GSTIN."*
- [ ] Enter a valid one (29ABCDE1234F1Z5) → accepted, stored upper-case
- [ ] Currency is fixed at INR and cannot be edited
- [ ] Toggle **Prices include tax** → the explanation below it changes
- [ ] Save, reload → the toggle held its new position

#### Tax

- [ ] Five GST slabs are listed: 0, 5, 12, 18, 28%
- [ ] **GST 18%** is marked Default
- [ ] Rates display as 18.00%
- [ ] Add a rate named **GST 3%** at 3 → appears in the list
- [ ] Add one with the same name → refused as a duplicate
- [ ] Deactivate a non-default rate → shows Inactive
- [ ] Try to deactivate the **default** rate → refused, with the reason

#### Payments

- [ ] Cash, UPI, Card, Bank Transfer are listed with their type badge
- [ ] Cash carries a **Cash drawer** badge; the others do not
- [ ] Deactivate UPI → shows Inactive; reactivate it

- [ ] Try to deactivate **Cash** → refused: at least one cash method must stay active

## Expenses

- [ ] Seven categories: Rent, Electricity, Salaries, Transport, Packaging, Repairs, Miscellaneous
- [ ] Add one → appears immediately
- [ ] Add the same name again → refused

## 2. Branches

**Settings** → **Branches** as admin.

- [ ] MAIN and NORTH are listed with manager and user counts
- [ ] **Add branch** — code **MGROAD**, name **MG Road** → created and listed
- [ ] The code is forced to upper case even if typed lower
- [ ] Add another with code **MGROAD** → *"Branch code MGROAD is already in use."*
- [ ] A code containing a space or ! is rejected inline
- [ ] Open the new branch, change its name, save → the list shows the new name
- [ ] Assign a manager from the dropdown → shows in the Manager column
- [ ] Leave the manager empty → **saves without error** (*this failed silently before — see §5*)

### Deactivation — the rule that matters (FR-3.8)

- [ ] Deactivate **MG Road** → badge turns Inactive, **the row stays**
- [ ] Open the header **branch switcher** → MG Road is **gone**
- [ ] Reactivate it → it comes back in the switcher
- [ ] Deactivate every branch but one, then try the last → refused: at least one branch must stay active

## 3. Customers and suppliers

**Customers** in the sidebar.

- [ ] The list is empty, or holds only what you have created
- [ ] **Add customer** with a name only → saves. A walk-in has nothing else
- [ ] Add one with phone **9876543210**
- [ ] Add a second with the **same phone** → *"Another customer (...) already uses this phone number."*
- [ ] Give a **supplier** that same phone → **allowed**, they are different directories
- [ ] Edit a customer, keep their phone unchanged, save → no false duplicate
- [ ] Enter **not-a-phone** → rejected inline
- [ ] Enter **MiXeD@Example.COM** → stored lower-case (check in Studio)

### Search and status

- [ ] Search by part of a name → matches
- [ ] Search by phone → matches
- [ ] Search by email → matches

- [ ] Search by GST number → matches
- [ ] **Deactivate** a customer → disappears from the default list
- [ ] Click **Show inactive** → they reappear, badged Inactive
- [ ] Reactivate → back to normal

Repeat the same pass on **Suppliers**. Suppliers additionally have a **Company** field; customers do not.

## 4. Permissions

| User    | Should see in the sidebar                                                       |
|---------|---------------------------------------------------------------------------------|
| admin   | Customers, Suppliers, Business, Branches, Users, Roles, Audit log               |
| manager | Customers, Suppliers, Business, Users, Audit log — <b>no Branches, no Roles</b> |
| staff   | Customers, Suppliers — <b>no Settings group at all</b>                          |

- [ ] Each user's sidebar matches
- [ ] As **staff**, open </settings/business> directly → redirected
- [ ] As **staff**, open </settings/branches> directly → redirected
- [ ] As **staff**, open </customers/new> directly → redirected (they may view customers, not create them)
- [ ] As **manager**, </settings/branches> → redirected. Managers run a branch; they do not create them

At the API, signed in as staff (see M0 §4 for how to get the cookie):

```
curl -s -o /dev/null -w '%{http_code}\n' localhost:3000/api/branches?manage=1 \
-H "Cookie: $COOKIE"
curl -s -o /dev/null -w '%{http_code}\n' -X POST localhost:3000/api/customers \
-H "Cookie: $COOKIE" -H 'content-type: application/json' -d '{"name":"Sneaky}"'
```

- [ ] The POST returns **403**

## 5. Form errors are never silent

A bug found during M1: a validation failure with no visible message meant the Save button appeared to do nothing.

- [ ] Paste 300 characters into any address field → a visible error appears
- [ ] On every form, an invalid field shows red text under **that** field
- [ ] No form ever fails with the button simply doing nothing

## 6. Audit

**Settings** → **Audit log** as admin.

- [ ] Everything above is recorded: branch created, customer created, tax rate added, payment method deactivated
- [ ] Entity types include [branch](#), [customer](#), [supplier](#), [tax\\_rate](#), [payment\\_method](#), [business](#)
- [ ] Expanding a row shows old → new values



## 7. Responsive

At ~375px, ~810px and full width:

- [ ] Customers, Suppliers and Branches lists become **cards** below 768px
- [ ] Business settings tabs scroll rather than wrap
- [ ] Forms stack to one column; buttons go full width
- [ ] Nothing scrolls sideways
- [ ] The sidebar's **Master data** and **Settings** groups appear in the mobile drawer

## What M1 deliberately does not include

- **No customer/supplier history tab yet.** Clicking a customer opens the edit form. The profile view showing their purchases, returns and credit arrives with M4/M5, when there is something to show. *This is a known gap against M1's stated criteria, carried into M2.*
- No products, stock or IMEI's — that is M2
- Attachments have a working API and storage layer, but **no upload button on any screen yet**; the first screen that needs one is M3's purchase form
- Logo upload is not wired to the business profile

## M3 — Purchases · specs on a line

*Added after M11, when a real shipment showed the gap: a purchase knew the product and the IMEI and nothing else, so ten iPhones arrived as ten handsets that did not know they were 256GB green — and each one had to be opened and edited afterwards.*

- [ ] On a **serialised** line, **Specs for this line** shows RAM, Storage, Colour and Variant
- [ ] They are **not** shown on a counted line — a box of cables has no RAM
- [ ] Book in 2 handsets with RAM 8 GB, Storage 256 GB, Colour Green
- [ ] The purchase page shows 256 GB · 8 GB · Green under the product name. *A bill for "iPhone 17 × 10" is ambiguous a month later*
- [ ] Open each handset in Devices — **both** carry all three
- [ ] Book in 3 more; on the second only, open its spec box (the slider icon beside the IMEI) and set Colour = Black, battery = 87
- [ ] That handset is Black at 87%; the other two are Green with no battery figure
- [ ] The odd one still shows **256 GB** and **8 GB**. *Blank means "same as the line", not "empty"*
- [ ] Scanning still works: type an IMEI, press Enter, and the cursor lands on the **next IMEI box** — never in a spec field
- [ ] **Duplicate line** copies the product, cost and specs and leaves the IMEI's blank. *A mixed shipment is the same phone in four storages*
- [ ] A handset booked in before this change still opens and edits normally

## What this does not do

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- **No specs on the product.** "iPhone 17 256GB Green" as its own product would be 24 products for one model, invented at the goods-inwards desk in whatever spelling was to hand — and used stock would still differ piece by piece
- **No retro-fill.** Changing a confirmed line's specs does not rewrite handsets already created; correct those on their own screens

## Lists: paging and sorting

*Carried debt, cleared after M11. These four screens had no page control at all: a shop with sixty users saw twenty-five and no sign there were more.*

- [ ] **Users, Branches** and **Low stock** each say **Showing 1–25 of N**, or explain themselves when empty
- [ ] Click a column heading — the rows re-order and the URL gains **sort=** and **dir=**
- [ ] Click the same heading again — it reverses
- [ ] Copy that URL into another tab: the same sorted view comes back. *Sorting is a link, not something the page remembers privately*
- [ ] Go to page 2, then sort — you land back on **page 1**. *Staying on page 7 of a re-ordered list shows a stranger's rows*
- [ ] On Low stock, the default order is **worst first** — furthest below its minimum at the top, not alphabetical
- [ ] Sort Low stock by Product, then page forward: the second page continues the alphabet. *It is sorting the whole list, not the page on screen*
- [ ] A blank cell (a branch with no city, a user who has never signed in) sorts to the **bottom** either way
- [ ] **Roles** shows a count and has **no** page control. *It is an editor: paging the list would mean turning a page to reach the role you came to edit*
- [ ] Edit **?sort=** in the address bar to something invented — the screen falls back to its normal order rather than breaking

**On a phone**, where the tables become cards:

- [ ] A **Sort** strip sits above the cards, scrolling sideways
- [ ] Tapping one re-orders the cards and marks it as the active sort
- [ ] Tapping it again reverses. *Sorting must exist on the screen the owner actually carries*

## M4 — Sales & Billing

**Delivers.** PRD FR-6.1 – FR-6.8, FR-26.1, FR-26.3, FR-26.4, FR-38.2, FR-38.3.

This is the screen the shop lives on. Everything before it existed so that this one could work: a bill is built, money is taken, stock drops, and an invoice comes out. Test it standing up, at speed, the way the counter will.

**Before you start.** You need stock. Run through M3 once to bring in at least one USED or ER handset with an IMEI, and one accessory with a quantity of ten.

### 1. Speed at the counter

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Go to **Billing**.

- [ ] The search box already has focus — you can start typing without clicking

- [ ] Type an accessory name → results appear as you type
- [ ] With exactly one result, pressing **Enter** puts it on the bill
- [ ] Scan or paste an IMEI → that exact handset lands on the bill
- [ ] **Time yourself:** accessory → Cash → Save, keyboard only, under 20 seconds
- [ ] The whole flow needs no mouse

## 2. The bill

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- [ ] A device line shows the IMEI or serial underneath the product name
- [ ] A device line shows its main type (USED / ER / ACT / GLOBAL)
- [ ] A GLOBAL device also shows **NEW CUT** where that applies
- [ ] A device line's quantity is fixed at 1 and cannot be edited
- [ ] An accessory line's quantity can be raised, and the total follows
- [ ] Adding the same IMEI twice is refused with a clear message
- [ ] Line discount reduces that line only; bill discount reduces the total
- [ ] **Staff** signed in cannot give a discount at all
- [ ] Totals: taxable + GST equals the total, exactly, to the paisa
- [ ] Change a quantity and the totals update instantly

## 3. The bill survives real life

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- [ ] Build a half-finished bill and **reload the page** → it is still there
- [ ] Clear bill empties it, and asks nothing you would regret confirming
- [ ] Switch branch with a bill open → the bill clears (it belonged to the other branch's stock)

## 4. Payment

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- [ ] **+ Cash** fills the full balance in one press
- [ ] Split across two methods (say Cash 500 + UPI rest) → both are recorded
- [ ] Pay less than the total with **no customer attached** → refused, because an unpaid balance needs someone to owe it
- [ ] Attach a customer, then save partly paid → the sale shows **PARTIAL**
- [ ] Overpaying is refused

## 5. Customer, without leaving the bill

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- [ ] Press **New** beside the Customer box
- [ ] Name and phone only → **Add to bill**
- [ ] The new customer is created *and* attached, and the bill is untouched
- [ ] A duplicate phone number is refused with a message naming the clash

- [ ] Cancel closes the dialog and changes nothing

## 5c. Pickers do not silently truncate

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Customers, suppliers and products are chosen through a **searchable picker**, never a fixed dropdown — a capped list drops records with no hint that it did.

- [ ] The Customer box on billing opens a search, not a long dropdown
- [ ] Typing a phone number finds the customer
- [ ] A customer created minutes ago is findable
- [ ] Same for **Supplier** on the purchase form
- [ ] Same for **Product** on the purchase form and on Add device

## 5b. Customer history (FR-6.7)

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Open **Customers** and click a customer who has bought something.

- [ ] The **History** tab opens first — that is what you came to see
- [ ] Total spend, number of purchases, outstanding and last purchase date
- [ ] Every invoice is listed, newest first, with branch and item count
- [ ] Clicking an invoice number opens that sale
- [ ] Purchases from **every branch** appear, not just the current one
- [ ] A part-paid bill shows **PARTIAL**, and outstanding is non-zero and red
- [ ] A voided bill is listed but does **not** count towards spend or dues
- [ ] A customer who has bought nothing says so, rather than showing an empty table
- [ ] The **Details** tab still edits the customer and saves
- [ ] On a phone the history becomes cards and nothing overflows sideways

## 6. The invoice

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Save a bill; you land on the sale.

- [ ] The invoice shows shop name, GSTIN, branch, invoice number and date
- [ ] Customer details appear, or "Walk-in customer"
- [ ] Every line shows quantity, price, tax and amount
- [ ] GST is broken out per rate
- [ ] **Print A4** produces a clean full-page invoice with no navigation on it
- [ ] **Print receipt (80 mm)** produces a narrow slip that fits the thermal roll
- [ ] **Download PDF** saves a file named after the invoice number
- [ ] Open the PDF: every figure matches the screen, and no character shows as a blank box
- [ ] The PDF's amounts are labelled INR (the rupee glyph is deliberately not used — see <src/server/pdf/invoice-pdf.tsx>)

- ☐ **Share** hands the PDF to the phone's share sheet (WhatsApp, mail); on a desktop browser it copies the invoice link instead

## 6b. The invoice is a statutory GST document (OQ-4)

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First set **Settings** → **Business** → **GST state**, and a **GST state** on each branch. Give a few products an **HSN code** (8517 phones, 8544 cables).

- ☐ The invoice is headed **Tax Invoice**
- ☐ **Place of supply** appears, with the state code and name
- ☐ Each line shows its **HSN code**
- ☐ An ordinary local sale shows **CGST** and **SGST** as separate lines, each half the rate — never a single combined "GST 18%"
- ☐ CGST + SGST adds up to exactly the tax shown
- ☐ An **HSN summary** table appears below the totals on A4
- ☐ The HSN summary's taxable and tax columns add up to the invoice totals
- ☐ Set a customer's **GST state** to another state, bill them → the invoice shows a single **IGST** line and no CGST/SGST
- ☐ The customer pays the same total either way — only the split changes
- ☐ A product with no HSN still bills; the line shows "—" and stays out of the summary
- ☐ The downloaded PDF matches the screen on every one of these

## 7. Stock actually moved

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- ☐ The handset you sold is now **Sold** on the device page
- ☐ Its IMEI no longer appears in billing search
- ☐ The accessory's branch stock dropped by the quantity sold
- ☐ The stock ledger shows one **SALE** row referencing the invoice

## 8. The other billing system (FR-38.2, FR-38.3)

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- ☐ A **NEW** handset never appears in billing search at all
- ☐ Forcing one onto a bill is refused server-side too
- ☐ Open that device → **Sold in other system** is offered
- ☐ Mark it → status becomes **SOLD\_PENDING\_IMPORT** and stock drops now
- ☐ It disappears from billing search, so it cannot be sold twice
- ☐ An ECITY-channel device does **not** offer that button

## 9. Sales list

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- ☐ Every saved sale is listed, newest first
- ☐ Search by invoice number, customer name or IMEI finds it

- [ ] Payment status filter: PAID / PARTIAL / UNPAID each return the right rows
- [ ] Branch filter narrows to that branch; a branch-scoped user sees only theirs
- [ ] From/To dates include both end days
- [ ] Filters survive a page reload (they are in the URL, so they are shareable)

## 10. Concurrency — needs two browsers

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- [ ] Sign in on two windows, put the **same IMEI** on both bills
- [ ] Save both → the first succeeds, the second fails with a clear message
- [ ] The device is Sold exactly once, and only one sale exists
- [ ] Save a bill, then press Save again on a restored copy → still one sale

## 11. Responsive

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- [ ] On a phone the billing screen is usable one-handed; nothing overflows sideways
- [ ] The cart and the totals panel stack rather than squeeze
- [ ] The invoice is readable on a phone
- [ ] On a large screen the search, cart and payment panel sit side by side

## What M4 deliberately does not include

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- **No returns or exchanges** — that is M6
  - **No trade-in** on the bill — also M6
  - **No Excel import** of the other system's sales — that was M12, now deferred because the two businesses are separated (docs/02 §2.3); devices marked `SOLD_PENDING_IMPORT` are waiting for it
  - **Returns do not appear in customer history yet** — the tab says so; they arrive with M6. Payments now have their own Statement and Receipts tabs
  - No daily cash-up or Z-report — M7
- 

## Cross-cutting — Lists & pagination

Every list in the app is paginated server-side at **25 rows a page**. Check these on **Products, Devices, Customers, Suppliers, Purchases, Sales** and the **Audit log** — the control is the same one everywhere.

- [ ] Under each list: *"Showing 1–25 of N"*, with the real total
- [ ] A list with more than 25 rows offers **Previous / Next**
- [ ] A list with 25 or fewer offers no page buttons, but still states the count
- [ ] On page 1, **Previous** is greyed out and does nothing when clicked
- [ ] On the last page, **Next** is greyed out
- [ ] **Next** advances, and the rows are genuinely different
- [ ] Page 1's URL has no `?page=` on it — one address for the first page
- [ ] Apply a filter, then page forward → **the filter is still applied**

- [ ] Change a filter while on page 3 → you go back to page 1, not an empty page
- [ ] Copy a page-2 URL into a new tab → same page, same filters
- [ ] Browser back returns to the previous page of results
- [ ] On a phone the buttons are big enough to tap and do not overflow sideways

## M4 — Billing · price prefill and camera scanning

*Added after M13. The till showed a blank price on every handset; and the counter has a laser scanner but a salesperson on the floor has only a phone.*

### Price

- [ ] A purchase line has **Selling price (₹)** beside the unit cost
- [ ] Book a handset in at cost 18,000, selling 24,000
- [ ] Scan it at the till — the price shows **24,000** without typing
- [ ] Book another in with the selling price **left blank**
- [ ] At the till it shows the **product's list price**. *Blank means "use the list price", not "free"*
- [ ] A handset bought before this change also prefills now, from the product
- [ ] Change the price on the bill — it still saves what you typed

### Camera (on a phone)

- [ ] On a **desktop with no camera**, no camera button appears
- [ ] On a **phone**, a camera button sits beside the scan box
- [ ] Tap it — the back camera opens with a frame to aim through
- [ ] Point at an IMEI barcode: it fills the box and closes by itself
- [ ] The item goes on the bill exactly as if you had typed it
- [ ] Scan the **serial** barcode next to it — it does *not* auto-add; it shows what it read and asks. *Putting the wrong handset on a bill is worse than a slow scan*
- [ ] Deny camera permission — it says so plainly instead of hanging
- [ ] Over plain **http** it explains that the camera needs a secure address
- [ ] The purchase form's IMEI boxes each have the same camera button
- [ ] The **laser scanner still works** on the desktop till, unchanged

## M5 — Payments, Credit Sales & Customer Dues

**Delivers.** PRD FR-7.1 – FR-7.6.

Credit is where a shop quietly loses money. This module exists so that the answer to "who owes us what, and since when?" is one screen and always right.

**Before you start.** Set **Settings** → **Business** → **Default credit period** (30 days is the default). You need one product in stock and one customer.

## 1. A bill can leave the counter unpaid

---

- [ ] Bill something to a **named customer** and pay nothing → the **Credit terms** panel appears, with a due date already filled in
- [ ] Pay the bill in full → the credit panel disappears (a cash sale needs no due date)
- [ ] Pay part of it → the panel comes back
- [ ] Try to leave a balance with **no customer** → refused, because a walk-in cannot be given credit
- [ ] Change the due date by hand and save → the sale shows that date
- [ ] Save unpaid → the sale shows **UNPAID**; part-paid shows **PARTIAL**
- [ ] Open that sale: it shows **Outstanding**, the **due date**, your credit note, and a **Collect payment** button
- [ ] Past the due date, the sale says *overdue* in red
- [ ] A fully paid sale shows none of that panel

## 2. Customer dues (FR-7.6)

---

Open **Sell** → **Customer dues**.

- [ ] Your unpaid customer is listed, with the amount owed
- [ ] **Total outstanding** matches the sum of the rows
- [ ] The **By age** panel adds up to the same total
- [ ] A bill created today sits in the **0–7 days** column
- [ ] Search by customer name and by phone number
- [ ] Filter by branch → only that branch's bills count
- [ ] **Overdue only** hides a bill whose due date has not passed yet
- [ ] A customer who has paid everything drops off the list entirely
- [ ] On a phone the rows become cards and nothing scrolls sideways

## 3. Collecting money (FR-7.3)

---

- [ ] Press **Collect** on a row → the customer's open bills are listed, oldest first
- [ ] Type an amount → it is allocated down the list, oldest first, and you can see exactly which bills it lands on before saving
- [ ] **Settle everything** fills in the exact total owed
- [ ] **Choose invoices** lets you allocate by hand instead
- [ ] Allocating more than a bill owes is refused, and says which bill
- [ ] Save → you land on a printable receipt with its own number
- [ ] The bill is now **PAID** on the sales list *and* on the sale itself
- [ ] Three part payments settle a bill and flip it to Paid on the third



## 4. Paying at a different branch (FR-7.5)

---

- ☐ Bill something on credit at **Branch A**
- ☐ Collect the money at **Branch B** (change *Collected at* on the form)
- ☐ The receipt shows Branch B; the invoice still shows Branch A
- ☐ The statement shows both, each against its own line

### 4b. Branch-wise (FR-7.6)

---

The **By branch** panel appears on the dues screen when there is more than one branch with activity.

- ☐ Each branch shows outstanding, overdue, collected and receipt count
- ☐ The **All branches** row adds up the columns above it
- ☐ Outstanding here matches the total at the top of the page
- ☐ Bill on credit at Branch A, collect at Branch B → A's *outstanding* falls, B's *collected* rises. They are not supposed to match
- ☐ Change the **Collected from / to** dates → only the collected column moves; what is owed is owed regardless of the window
- ☐ Void a receipt → it stops counting as collected, and the debt returns to the branch that raised the bill

## 5. The receipt (FR-26.2)

---

- ☐ Shop name, branch, receipt number, date, customer
- ☐ The amount, the method, and the reference if you gave one
- ☐ **Applied to** lists each invoice and how much went to it
- ☐ Money beyond what was owed shows as **held on account as an advance**
- ☐ **Balance after this receipt** matches the customer's account
- ☐ Print A4 and Print receipt (80 mm) both produce clean output

## 6. Voiding a receipt

---

- ☐ **Void** is offered to the owner and manager
- ☐ It is **not** offered to counter staff — they may collect, not erase
- ☐ Voiding needs a reason
- ☐ After voiding: the receipt is marked VOIDED but still on record
- ☐ The invoice it settled is **UNPAID** again
- ☐ The customer's balance has gone back up
- ☐ The statement shows a **Reversal** line — the old line is untouched
- ☐ Voiding the same receipt twice is refused

## 7. The statement (FR-7.6)

---

Open a customer → **Statement**.

- ☐ Every movement, oldest first, with a running balance
- ☐ An invoice appears as a debit; a payment as a credit
- ☐ Clicking an invoice number opens that sale; a receipt opens that receipt
- ☐ The closing balance matches the figure at the top of the customer page
- ☐ A customer in credit shows a negative balance labelled *in credit*

## 8. The numbers reconcile

---

This is the one that matters. Pick any customer with dues:

- ☐ Add up what their open invoices still owe, by hand
- ☐ Compare with the balance on the dues list — they must be identical
- ☐ Compare with the statement's closing balance — also identical
- ☐ Void a receipt and repeat: all three still agree

## What M5 deliberately does not include

---

- **No returns or exchanges** — M6. A refund is not a negative payment
  - **No cash drawer or daily closing** — M7. Money collected is recorded, but the day is not yet reconciled
  - **No statements by email or WhatsApp** — M13
  - **No interest, credit limits or dunning letters** — not in v1 at all
- 

## M6 — Returns, Exchange & Trade-In

**Delivers.** PRD FR-8.1 – FR-8.5, FR-9.1 – FR-9.3, plus correcting a device.

Goods come back and old phones come in. The rule underneath everything here: **a returned handset is never immediately sellable again.**

**Before you start.** You need a completed sale with a handset on it and another with an accessory. Sign in as **admin** unless a step says otherwise.

### 1. Taking a return (FR-8.1)

---

- ☐ **Sell** → **Returns** → **Take a return** offers one search box
- ☐ Find a bill by its **invoice number**
- ☐ Find the same bill by **customer name**
- ☐ Find it by the **IMEI** of a handset on it
- ☐ Opening a sale and pressing **Take a return** gets you to the same place
- ☐ Only lines with something left to return are offered

- [ ] Returning 2 of 5 gives back exactly two fifths of that line
- [ ] Asking for more than remains is refused, and says how many are left
- [ ] Return 3 of 4, then try to return 2 more → refused, 1 left
- [ ] A whole bill returned at once is marked **FULL**; part of one is **PARTIAL**
- [ ] Goods can be returned to a **different branch** than they were sold at

## 2. A returned handset is not sellable (FR-8.2)

---

This is the one most likely to be got wrong. Test it properly.

- [ ] Return a handset → it appears in **Returns** → **Inspection queue**
- [ ] Search that IMEI at the till → **nothing comes up**
- [ ] The device page shows status **RETURNED**
- [ ] An accessory behaves differently: it goes straight back into branch stock and can be sold again immediately

## 3. Inspection (FR-8.3)

---

- [ ] **Inspect** offers Available, Used, Damaged, Repair required
- [ ] Grading **Available** → the handset leaves the queue and reaches the till
- [ ] Grading **Used** → also sellable, and the condition is recorded
- [ ] Grading **Damaged** → status DAMAGED, still not sellable
- [ ] Grading **Repair required** → status REPAIR, still not sellable
- [ ] Notes typed during inspection appear on the device
- [ ] Inspecting something that is not awaiting inspection is refused
- [ ] Signed in as **staff**: the queue is visible but there is **no Inspect button** — releasing stock for sale is not the counter's decision

## 4. Classification survives the whole path (FR-8.4)

---

The acceptance test for this module. Do it exactly.

- [ ] Buy a handset as **GLOBAL** with **NEW CUT** ticked
- [ ] Sell it, then return it
- [ ] In the inspection queue it still shows **GLOBAL**
- [ ] Grade it **Used**
- [ ] On the device page it **still reads GLOBAL and NEW CUT**, with condition Used recorded separately

Condition and classification are different facts. Grading a handset's condition must never rewrite what kind of device it is.

## 5. The money (FR-8.5)

---

- [ ] Refund to a **payment method** — money goes back out

- [ ] Refund to the **customer's account** — no cash moves, they owe less
- [ ] A walk-in cannot have their account credited; it is refused
- [ ] A **deduction** (restocking fee) reduces what is handed back
- [ ] **The one that matters:** a customer who paid in full, returned the goods and got cash back ends with a **zero balance** — not in credit. Check Customers → their Statement
- [ ] Returning against an **unpaid** bill just reduces what they owe

## 6. Trade-in (FR-9.1 – FR-9.3)

---

On the billing screen, with something in the cart:

- [ ] **Trade-in** → **Add** takes IMEI, type, storage, colour, battery, condition notes, estimated value and agreed value
- [ ] Accepting it adds the old handset to **this branch's stock** immediately
- [ ] Its device page shows the type you chose, including GLOBAL/NEW CUT
- [ ] Its history starts with a **PURCHASED** event
- [ ] Its purchase price is the **agreed value**
- [ ] It is marked as sold here, never the other billing system
- [ ] On the bill: **Bill total** stays the full price, and the trade-in shows as a deduction against what is still to pay
- [ ] Save the bill → the difference is what the customer owed
- [ ] The invoice still shows the full price and full GST — a trade-in is not a discount, and the trade-in appears under the total as settlement, beside the cash, with the old handset's IMEI
- [ ] **Take one in for the full price of the phone and pay nothing.** The bill saves, reads **PAID**, and shows no outstanding balance and no due date. *A bill settled with the handset must not leave a receivable to chase*
- [ ] Do the same for a **named customer**: their statement shows the sale, the trade-in against it, and a balance of zero
- [ ] The sale list and the customer dues screen both read that bill as PAID — settlement is answered in one place, so they cannot disagree
- [ ] Trading in with **no customer** on the bill is allowed when the handset covers the whole total; it is refused only when something is left owing

### 6b. Refunding no more than was taken

---

- [ ] Sell on **credit** to a customer, pay nothing, then return the goods and choose to refund to **Cash** → the refund is ₹0 and their balance goes to zero. *They never paid, so nothing goes back out*
- [ ] Sell ₹1,000, pay ₹400, return everything, refund to Cash → ₹400 goes back and the remaining ₹600 comes off their account
- [ ] Same unpaid credit sale, but refund to the **customer's account** → the full value is credited. *A credit note is not money leaving the till*
- [ ] Return the rest of a part-returned bill → the two refunds together never exceed what was paid

## 6c. The GST switch (FR-2.6, FR-26.6)

**Settings** → **Business** → **GST**. Turn *Registered for GST* off and save.

- ☐ The **GST number** and **GST state** fields disappear from business settings, and so does the **Tax** tab and **Prices include tax**
- ☐ A new product form has no **HSN code** and no **Tax rate**
- ☐ A new customer, supplier and branch form has no **GST number**
- ☐ A device form has no **Tax rate**
- ☐ At the till, the total shows **Bill total** only — no Taxable, no Tax
- ☐ Sell a phone at ₹10,000 and take ₹10,000 → the bill total is exactly ₹10,000. *The customer pays the price on the label*
- ☐ Its invoice is headed **Invoice**, not Tax Invoice, and shows no GSTIN, no HSN column, no tax column and no HSN summary
- ☐ Download the PDF — same on paper as on screen

**Now turn GST back on and save.**

- ☐ Reopen that bill. It **still** reads Invoice, with no GSTIN and no tax. *This is the important one: nothing stores the PDF, so every reprint is re-rendered. A bill issued before the shop registered must never reprint as a tax invoice*
- ☐ Sell another phone → this one **is** a Tax Invoice, with GSTIN, HSN and the tax breakdown
- ☐ Turn GST off again and reopen that taxed bill → it still shows its GST. *It works in both directions*
- ☐ Turn GST back on before carrying on with the rest of the checklist

## 7. Correcting a device

- ☐ A device page shows **Edit** for admin and manager
- ☐ Signed in as **staff** there is no Edit button
- ☐ Change the **main type** and save
- ☐ The device history shows a **RECLASSIFIED** entry with what changed — corrections are recorded, not silent
- ☐ Change **Billed in** from the other system to ECITY → that handset now appears at the till. *This is the fix for NEW stock that was invisible*
- ☐ NEW CUT on anything that is not GLOBAL is refused
- ☐ A **sold** device cannot be edited — an issued invoice describes it
- ☐ The IMEI is shown but not editable

## What M6 deliberately does not include

- **No voiding a return.** Take a new sale if goods go back out
- **No repair workflow** — REPAIR is a status, not a job card
- **No automatic valuation** of trade-ins; the shop decides the number
- **No returns against the other billing system's sales** — those bills do not exist here at all — NEW stock belongs to the other business (docs/02 §2.3)

## Housekeeping

The end-to-end tests create real branches and customers in your development database. After a test run you will see extra rows. To get back to a clean state:

```
dropdb ecity && createdb ecity
psql -d ecity -c "CREATE ROLE ecity WITH LOGIN PASSWORD 'ecity' SUPERUSER;"
npm run db:migrate && npm run db:seed
```

## Sign-off

| Tested by      |             |
|----------------|-------------|
| Date           |             |
| Build / commit |             |
| Result         | pass / fail |
| Issues raised  |             |

## M7 — Expenses, Cash Drawer, Accounts & Daily Closing

Sign in as **admin** unless a step says otherwise. Work a whole day through in one sitting — the point of this module is that the day balances at the end.

### 1. The drawer sees every rupee (FR-11.2)

Open **Cash drawer** and note *Expected in the till*. After each step below, come back and check it moved the right way and by the right amount.

- [ ] Bill something for **cash** → expected goes **up**
- [ ] Bill something by **UPI** → expected does **not** move. *UPI never touches the till*
- [ ] Collect a customer's outstanding balance in **cash** → **up**
- [ ] Record a **cash expense** → **down**
- [ ] Pay a supplier in **cash** → **down**
- [ ] Refund a return in **cash** → **down**
- [ ] Every one of those appears as its own line under the total, with a time
- [ ] Switch branch → the other branch's till is a separate figure. *One branch's takings must never appear in another's*

### 2. Expenses (FR-10.1 – FR-10.3)

- [ ] **Record an expense** takes branch, category, amount, date, how it was paid, description and reference
- [ ] Choosing a non-cash method offers an **account**; choosing cash does not

- [ ] It appears in the list, newest first, with who recorded it
- [ ] Click the category → the expense opens on its own page
- [ ] **Attach a receipt** — a photo or PDF — and it shows on the expense
- [ ] A voided expense offers no upload
- [ ] **Total spent** is over everything in the filter, not just this page
- [ ] **Void** one → it asks why, and refuses an empty reason
- [ ] After voiding, the money is back in the drawer and the expense is gone from the live list. *It is voided, not deleted — tick "include voided" and it is still there with its reason*

### 3. Accounts (FR-12.1 – FR-12.4)

---

- [ ] **Add account** for a bank and one for UPI, with an opening balance
- [ ] An account can belong to one branch or to **all branches**
- [ ] **Transfer** between them → both balances move, by the same amount
- [ ] Click an account name → its **ledger**: every movement, newest first, with a running balance you can prove the total from
- [ ] The two legs of the transfer appear, one in each account
- [ ] **Reconcile** one, entering a statement balance ₹500 *below* the books
- [ ] The balance does **not** change to match. The ₹500 shows as **unreconciled**. *A real difference wants an explanation, not an automatic adjustment*

### 4. Closing the day (FR-13.1 – FR-13.4)

---

- [ ] **Daily closing** shows sales, credit given, collected, returns and refunds, what each payment method took, and the expenses by category
- [ ] It shows **invoices and items**. Sell three of one thing on one bill → one invoice, three items
- [ ] Type a counted amount ₹200 **below** expected → it says ₹200 **short** as you type, before you commit to anything
- [ ] Close the day → the shortage is recorded with the branch, your name and the time
- [ ] The drawer for that day now reads **CLOSED**

### 5. A closed day is never rewritten (OQ-5) — *the important one*

---

- [ ] Signed in as **staff**, try to record an expense dated into the closed day → refused
- [ ] As **admin**, record one → allowed, and the movement is tagged **After close**
- [ ] Go back to the closing screen. It says **this day was corrected after it was closed**, and shows both figures: what was signed, and what the movements now come to
- [ ] The signed **expected, counted and difference are unchanged**. *This is the whole decision. If a closing could be edited, coming up short today and adjusting yesterday would make the difference disappear*

## 6. Reopening a day

---

- [ ] On a closed day, **Reopen this day** → asks why, and refuses an empty reason
- [ ] It reopens, and the day can be closed again
- [ ] Close a **later** day for the same branch, then try to reopen the earlier one → **refused**, and it tells you to post a correction instead
- [ ] **Closing history** shows the reopened day marked **Reopened**. *The voided closing is kept — that a day was closed and reopened is part of the record*

## 7. Two branches (FR-13.5)

---

- [ ] Close each branch's day separately
- [ ] **All branches** shows both side by side and the consolidated totals add up
- [ ] A branch that has not closed still appears, marked **Open**, with what it should have

## 8. Tomorrow starts from what was counted

---

- [ ] Close a day counting ₹100 **less** than expected
- [ ] Open the next day's drawer → its **opening** is what you counted, not what was expected. *The drawer starts with the cash actually in it, so a shortage does not repeat itself every morning*

## 9. Correcting a purchase (carried in from M5)

---

- [ ] A purchase page shows **Correct details** for admin and manager
- [ ] It offers the **supplier bill number, date and notes** — and nothing else
- [ ] There is **no** way to change a line, quantity or cost. *Stock has moved and the supplier ledger has been posted; reversal is the honest tool*
- [ ] Save a corrected bill number → it shows on the purchase
- [ ] Signed in as **staff**, there is no Correct details button

## What M7 does not do

---

- **No cash counting by denomination**; one counted total per day
- **The other business's sales are not in the expected figure**, by design (docs/02 §2.3). ECITY's closing covers what ECITY billed. **If both take cash into the same physical drawer, every day will read as an overage** — settle that operationally before go-live, with a separate drawer or a single cash movement recording the other takings

## M8 — Branch Transfers & Stock Adjustments

You need two branches with stock. Sign in as **admin** unless a step says otherwise.



## 1. Requesting (FR-3.6)

---

- [ ] **Transfers** → **Request a transfer**
- [ ] Picking a *From* branch shows only what **that branch actually holds** — handsets in stock and accessories with a count
- [ ] Changing the *From* branch clears what you had added. *Those items were on the other branch's shelf*
- [ ] A handset adds as one line with its IMEI; an accessory lets you type a quantity, capped at what is on hand
- [ ] Save → it appears as **REQUESTED**
- [ ] Nothing has moved yet: the handset is still sellable at the sending branch

## 2. Approving and sending

---

- [ ] The transfer list has **Waiting for approval** and **To receive** — the approval queue and the receiving screen, one click away
- [ ] The transfer offers **Approve**, then **Dispatch** — not both at once
- [ ] Signed in as **staff**, a requested transfer offers no Approve button. *Sending stock out of a branch is a manager's decision*
- [ ] As a manager assigned to the **receiving** branch only, the transfer offers no Approve or Dispatch — it says the other branch sends it
- [ ] Once in transit, that same manager **can** receive it, and a manager at the **sending** branch cannot. *Signing for a delivery is the receiving branch's job — whoever packed the box cannot attest it arrived*
- [ ] After **Dispatch** it reads **IN TRANSIT**

## 3. In transit belongs to nobody — the important one

---

- [ ] At the **sending** branch, search the IMEI at the till → **not offered**
- [ ] Switch to the **receiving** branch and search it → **not offered**
- [ ] The accessory count at the sending branch has **gone down** by what was sent
- [ ] The receiving branch's count has **not gone up** yet. *It is in a van. If it were sellable at both ends it would be sold twice*

## 4. Receiving (FR-3.7)

---

- [ ] The receiving screen offers a **scan box**, focused and ready
- [ ] Nothing is ticked to begin with. *A box you scan into is safer than one you untick — a handset the scanner misses shows as short instead of passing quietly*
- [ ] Scan (or type and press Enter) each IMEI → its line ticks
- [ ] A handset that will not scan can still be ticked by hand
- [ ] **Receive in full** → the transfer reads **RECEIVED**
- [ ] The exact IMEI is now at the receiving branch and sellable there
- [ ] Its device history shows **TRANSFERRED OUT** and **TRANSFERRED IN**, each naming **both branches** and the date. *Not one branch with the other hidden in the detail — M9's timeline reads these directly*

- [ ] Its **main type** and **NEW CUT** are unchanged. *A journey is not a reclassification*
- [ ] The accessory count at the receiving branch has gone up

## 5. Something did not arrive

---

- [ ] Send another handset and, on receipt, **untick** it
- [ ] A warning appears explaining it will be marked lost, and asks what happened
- [ ] After receiving, the transfer is flagged **Short on receipt** with your note
- [ ] The handset reads **LOST**, not still in transit. *It left one branch and reached no other — it has to be accounted for, not stranded*
- [ ] Try a partly received accessory line: send 5, receive 3 → only **3** arrive and the transfer is flagged short

## 5b. Something arrived that should not have

---

- [ ] On a receiving screen, scan an IMEI that is **not on the transfer**
- [ ] It is listed as unexpected, with a warning, and can be removed if you mis-scanned
- [ ] Receive → the transfer is flagged, and the stray IMEI is named in the discrepancy note
- [ ] That handset has **not** been moved to this branch. *Recording it is right; moving it would be inventing a transfer nobody authorised*

## 6. Cancelling

---

- [ ] Cancel a **requested** transfer → nothing moves anywhere
- [ ] Cancel one **in transit** → it asks why, and refuses an empty reason
- [ ] The handset is back at the **sending** branch and sellable again
- [ ] Its history shows the return **without** claiming it came from the other branch. *It never got there*
- [ ] Accessory stock is back at the sending branch too
- [ ] A **received** transfer offers no Cancel. *A completed movement is corrected by transferring back, not by rewinding it*

## 7. Adjustments (FR-28.1 – FR-28.3)

---

- [ ] **Adjustments** → **Adjust stock** asks first whether it is an accessory count or one handset
- [ ] For an accessory: pick it, type what you **actually counted**, and the change is shown as you type
- [ ] Entering the number the system already has is refused. *There is nothing to correct*
- [ ] Save with reason **Miscount** → the count changes and the entry shows **10 → 7** with who and when
- [ ] For a handset: **Miscount is not offered** — only Damage and Loss, and it explains that a wrong record is a device correction instead
- [ ] Adjust a handset as **Damage** → it leaves sellable stock and the till no longer offers it
- [ ] The adjustment lists the handset's **main type** and **NEW CUT as they were at the time**
- [ ] Click an adjustment → it opens on its own page

- [ ] **Attach evidence** — a photo of the damage or the count sheet
- [ ] The page says an adjustment is never edited, and that a wrong one is corrected by a second
- [ ] Signed in as **staff**, there is no Adjust stock button

## 8. Navigation

---

- [ ] **Transfers** and **Adjustments** appear under Inventory
- [ ] The transfer list filters by stage — **Requested** is the approval queue, **In transit** is the receiving one

## What M8 does not do

---

- **No printed transfer note** to travel with the goods
- **No partial dispatch**; a transfer goes in one movement or not at all
- **No printed barcode labels**; a scanner types the IMEI into the scan box like a keyboard, which is how these scanners work anyway
- **No transfer between businesses**, only between branches of one

# M9 — Global Search & IMEI Device History

The flagship. Sign in as **admin** unless a step says otherwise.

## 1. The search box (FR-30.1)

---

- [ ] A **Find anything** box sits in the header on every screen. *Deliberately not called "Search": every list has its own Search button that filters that list, and this does something different*
- [ ] **K** (or Ctrl+K) opens it from anywhere; *I* opens it too, unless you are already typing in a field
- [ ] **Escape** closes it
- [ ] On a phone it collapses to a magnifying-glass button
- [ ] Typing one character says to type more; nonsense says *Nothing matched*

## 2. Finding a handset (FR-30.5) — *the important one*

---

- [ ] Type a **full IMEI** → it opens that device page immediately, without a list. *A complete IMEI is unambiguous; making someone click a list of one is wasted work at a counter*
- [ ] Type the **first 10–12 digits** → a list of candidates, and you stay put
- [ ] Click one → its device page
- [ ] Register a **dual-SIM** handset (raise *IMEI fields per device* in Settings → Business first)
- [ ] Search its **second** IMEI → the same device page as the first. *A customer reads out whichever number is printed nearest*

## 3. The identity header

---

- [ ] The device page shows **main type** and **NEW CUT** where it applies

- [ ] **Every** identifier is listed, with the primary one marked
- [ ] A multi-IMEI handset says how many identifiers it has

## 4. The whole life (FR-30.6)

---

Take one handset through: purchase → transfer out → transfer in → sell on credit → return → inspect → correct something.

- [ ] The timeline shows **all seven** stages, oldest first
- [ ] Each entry is a **sentence**: *Sold on INV/... to Anil*, not just *SOLD*
- [ ] A transfer entry names **both branches**
- [ ] A reclassification says **what changed**, not just that something did
- [ ] Every entry with a document has a **working link** — click each one and land on that invoice, purchase, return or transfer
- [ ] **What it sold for** shows the price from the bill, the discount on that line, the customer, the cost and margin if you may see cost, the **payment status**, and **paid versus still owing**
- [ ] The Details card shows the **previous branch** for a handset that has moved, and a dash for one that never has
- [ ] A device sold, returned and sold again says so

## 5. Everything else searchable (FR-30.2 – FR-30.4)

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- [ ] **Customer name**, **phone**, **email** all find the customer
- [ ] **Supplier name** finds the supplier
- [ ] **Product name**, **SKU** and **barcode** find the product
- [ ] An **invoice number** finds the bill
- [ ] Results are **grouped** by kind, with devices showing their type

## 6. Permissions (FR-30.7)

---

- [ ] Signed in as the **branch manager** (MAIN only), search a customer who has only ever bought at **North** → **no results**
- [ ] A customer who has bought at **both** branches → found by both
- [ ] Search an IMEI sitting at **North** as that manager → no results
- [ ] Signed in as **staff**, searching still works for what they may see, and returns nothing they may not

## 7. Speed (PRD §9.1)

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- [ ] Search results appear in well under a second
- [ ] A device with a long history opens in **under 1.5 s**

## What M9 does not do

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- **No saved or recent searches**
- **No fuzzy spelling correction** — a mistyped name finds nothing
- **No search across voided documents** unless you open them directly
- **The timeline shows events, not field-level diffs** beyond naming what a reclassification changed; the audit log has the full before-and-after

## M10 — Dashboards & Analytics

Sign in as **admin** unless a step says otherwise. Do a day's trading first — a few bills, a return, an expense — so the numbers have something to say.

### 1. The dashboard (FR-15)

---

- [ ] **Today** shows sales, **purchases**, estimated profit, cash in the till, accounts, customer dues, supplier dues, stock value and returns
- [ ] Sales says how many bills and how many items
- [ ] Every figure is **clickable** and lands somewhere sensible
- [ ] The header's branch switcher changes the figures; "All branches" adds them up
- [ ] Signed in as **staff**, there is no estimated profit. *Profit needs cost prices, which staff do not see*

### 2. Alerts (FR-15.3)

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- [ ] Set a product's minimum above its stock → a **low stock** alert appears
- [ ] Leave a bill past its due date → an **overdue** alert
- [ ] Leave yesterday unclosed → an **unclosed day** alert
- [ ] Close a day with a shortage → a **cash mismatch** alert
- [ ] Each alert links to the screen that answers it
- [ ] With nothing wrong, it says **All clear**

### 3. Getting around the analytics

---

- [ ] **Analytics** in the sidebar opens the overview
- [ ] Eleven tabs: Overview, Insights, Sales, Products, Brands, Customers, Credit, Inventory, Profit, Payments, Suppliers
- [ ] Every page has From, To, a branch picker and a comparison button
- [ ] The presets — Today, 7 days, 30 days, 90 days, This year — set the range
- [ ] Every area shows a **chart and a table**
- [ ] **Compare to previous** appears everywhere except Inventory and Credit, whose headline figures are a position rather than a period
- [ ] Changing area **keeps** the range and branch

- [ ] The range is in the URL, so a view can be copied to someone else

## 4. The figures are right (criterion 1)

---

Pick a day you can add up by hand.

- [ ] **Sales**: revenue, bills, units and average bill match your arithmetic
- [ ] Revenue on the dashboard for today equals the Sales page for today
- [ ] **Profit**: revenue – cost of goods = gross; gross – expenses = estimated net
- [ ] **Payments**: what was taken at the counter plus what went on credit equals revenue
- [ ] **Brands** and **Products** add up to the same revenue as Sales

## 5. The branch selector (criterion 2)

---

- [ ] One branch, then the other → two different sets of figures
- [ ] Every branch → the two add up to the total
- [ ] Signed in as the **branch manager**, the other branch is not in the picker, and its figures never appear

## 6. Main type and NEW CUT (criterion 3)

---

- [ ] **Products** → **By main type** lists each type separately
- [ ] GLOBAL appears **twice**: once plain, once as **GLOBAL · NEW CUT**
- [ ] Their margins differ — which is the point of splitting them
- [ ] **Inventory** shows stock by type the same way
- [ ] **Profit** offers a main-type filter; **NEW CUT only** appears *after* choosing GLOBAL and disappears when you pick another type. *NEW CUT is a designation inside GLOBAL, never a sixth type*

## 6b. Insights (FR-35)

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- [ ] **Insights** shows revenue, average bill, margin and collection rate, each against the period before — with a direction, not just a figure
- [ ] **Which days earn** groups by weekday. *"Dead on Tuesdays" is actionable; "the 14th was slow" is not*
- [ ] Top products, brands and branches are listed
- [ ] The five main types are compared, GLOBAL split by NEW CUT
- [ ] Slow-moving stock is listed, each linking to its device
- [ ] Signed in as **staff**, margin is absent but everything else is there

## 6c. The rest of the measures

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- [ ] **Inventory** shows **turnover**, **low stock** and **out of stock**
- [ ] Set a product's minimum above its stock → it appears under *Needs reordering*; drop the stock to zero → it moves to out of stock

- [ ] **Credit** shows **collected in this period** and a **collection rate**, and lists customers **late more than once**
- [ ] A rate above 100% is not a bug — money arrives for older bills
- [ ] **Suppliers** shows *What we buy, and from whom*
- [ ] The **Overview** shows cash, accounts, customer dues and stock value beside the trade figures

## 7. Inventory movement (FR-21)

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- [ ] The movement table reads Opening → Purchases → Sales → Returns → Transfers → Adjustments → Current
- [ ] **Opening + in – out = Current.** *A movement report that does not reconcile is worse than none*
- [ ] **Handsets are counted, not just accessories.** Register a phone and sell it inside the period → purchases and sales each go up by one. *The stock ledger covers accessories; a phone shop's movement has to read the device history too*
- [ ] **Dead stock** lists handsets that were in stock before the period and are still here, oldest first

## 8. Reaching a handset (criterion 5)

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- [ ] From the dashboard, click **Stock value** → Inventory
- [ ] Click **Handsets** → the device list
- [ ] Click a device → its full history
- [ ] That is **three clicks** from the dashboard to one IMEI
- [ ] The other route also works: Overview → a branch → a bill → the IMEI on that bill opens the handset's history. *On a printed invoice the IMEI is plain text — paper has nowhere to click*

## 9. Speed (criterion 4)

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- [ ] Set the range to a whole year with every branch selected
- [ ] Every page answers in a couple of seconds at most

## What M10 does not do

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- **No nightly rollups.** Every figure is live from the documents. Fast enough at this size; revisit if the data grows
- **No exports yet** — that is M11
- **No automated recommendations** (PRD FR-35.4 puts them in a later version)
- **Charts are simple bars**, with no axes, tooltips or zoom
- **Estimated net profit is gross less recorded expenses.** Salaries and rent count only if somebody booked them as expenses

# M11 — Reports, Exports, Imports & Opening Balances

Sign in as **admin** unless a step says otherwise.

## 1. The report centre (FR-25.1 – FR-25.3)

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- [ ] **Reports** in the sidebar opens the centre
- [ ] Ten reports: Sales, Purchases, Inventory, Financial, Credit, Tax, Reconciliation, Branch comparison, Customers, Suppliers
- [ ] Each has From, To and a branch picker, and the filters are in the URL — copy the address to someone and they see the same figures
- [ ] The preview shows the first 100 rows and says how many there are
- [ ] Sales revenue for a period matches the Analytics page for the same period
- [ ] **Inventory** groups by main type, with **GLOBAL · NEW CUT** as its own value — never a sixth type
- [ ] **Credit** carries the 0–7, 8–30, 31–60 and over-60 buckets
- [ ] **Customers** and **Suppliers** list everyone with what they traded in the period and what is owed *now* — a list to call, and a list to pay
- [ ] Signed in as **staff**, the Financial report says it needs permission to see cost prices
- [ ] Changing the period says **Rebuilding...** while it works. *A report that looks frozen gets asked for three times*
- [ ] **Save this view**, name it, and it appears as a chip
- [ ] Go to another report, click the chip, and the first one comes back with its dates and branch
- [ ] Saving again under the same name **replaces** it rather than being refused
- [ ] Remove the view with its ×, and it goes

## 2. Exports (FR-33.3)

---

For at least Sales and Inventory:

- [ ] **CSV** downloads and opens in Excel with the accents intact
- [ ] **Excel** downloads as a real **.xlsx**, and the money column **adds up** when you select it. *A spreadsheet that cannot sum its own money column is not much use*
- [ ] **PDF** downloads, is landscape, and repeats the column headings on every page
- [ ] A very large PDF is refused with a message pointing at CSV, rather than silently cutting rows off

## 3. The catalogue (carried in from M5)

---

- [ ] **Settings** → **Catalogue** lists brands and categories with how many products each holds
- [ ] **Add a brand** → it appears, and shows in the product form's picker
- [ ] **Rename** it → the change shows everywhere
- [ ] **Deactivate** it → it stays listed, marked inactive. *Invoices already issued refer to it, so nothing is deleted*
- [ ] A duplicate name is refused
- [ ] **Add a category**, choosing whether it is tracked individually and by IMEI or serial
- [ ] On a category that **already has products**, changing how it is tracked is refused with a reason. *That would reinterpret stock that already exists*
- [ ] The page explains why **main types** are not managed here



## 4. The import wizard (FR-33.1)

Make a CSV with a heading row and a few customers, including one row with an empty name.

- [ ] **Import data** → choose Customers → pick the file
- [ ] It says how many rows it read, and **nothing has been created yet**
- [ ] The column mapping is **guessed** — "Full Name" finds *name*, "Mobile" finds *phone*
- [ ] The guess can be **changed**; a required field left unmapped is refused with a message naming it
- [ ] **Check the file** → it says how many are ready and how many have problems
- [ ] The bad row is named by its **line number as the spreadsheet shows it**
- [ ] **Download the full list of problems** gives a CSV of just those rows, with the row as uploaded beside the reason
- [ ] **Import** → only the good rows go in; the bad ones are left out and nothing is half-created
- [ ] The customers really exist afterwards
- [ ] Uploading the **same file again** is refused. *It would double everything in it*
- [ ] Signed in as **staff**, the import page redirects away

Then save the same list as **.xlsx** and do it again:

- [ ] The **.xlsx** uploads directly — no converting to CSV first
- [ ] An IMEI column typed as a number comes through as its digits, not as **3.81E+14**. *An IMEI that arrives in scientific notation matches nothing*
- [ ] A date cell arrives as a day, not as a timestamp
- [ ] An old **.xls** is refused with a message saying to save it as **.xlsx**

## 5. Handsets, with more than one IMEI (FR-33.2)

- [ ] Import a devices file with **IMEI No** and **IMEI 2** columns
- [ ] Both IMEIs land on the **same handset**, not two
- [ ] This works even while *IMEI fields per device* is set to 1 in Settings. *The file decides how many the handset has, not the form*
- [ ] A row with a main type that is not one of the five is rejected by name
- [ ] A row naming a product the shop does not have is rejected, and says so

## 6. Opening balances (FR-34.1 – FR-34.3)

**Settings** → **Opening balances**, three tabs.

- [ ] Declare **opening stock** for an accessory → the stock figure and the inventory report both show it
- [ ] Declare **opening cash** for a branch → the cash drawer shows it as the opening movement
- [ ] Declaring it twice for the same branch is refused
- [ ] Declare a customer's **existing due** → they appear on the dues screen, in the aging buckets, in the credit report and in the dashboard's "customers owe". *A balance nobody can see is worse than not importing it*
- [ ] Declare a supplier's **existing due** → it shows in supplier dues

- [ ] Every opening entry is dated as at the day you declared, not the day you typed it
- [ ] **Import a file instead** on the stock and dues tabs opens the wizard already set to the right kind
- [ ] A dues file naming somebody the shop does not have rejects **that row** and imports the rest. *A due against a name nobody recognises is worse than a missing row*
- [ ] A stock file uploaded without choosing a branch is refused straight away, not one row at a time

## What M11 does not do

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- **Only .xlsx, not the old binary .xls** — Excel saves one as the other in a click, and the message says so
- **Only the first sheet of a workbook** — several sheets is a question about which one, and guessing imports the wrong list
- **No saved report filters yet**; the URL carries them, which covers sharing but not naming
- **No scheduled or emailed reports**
- **No undo on an import**. A wrong batch is corrected the way anything else is — by voiding or adjusting what it created
- **OQ-9 is closed**: the go-live loading order, the checks after each step, and what is deliberately not migrated are in docs/04 §10

## M13 — Notifications, Alerts & Warranty

**Delivers.** PRD FR-27.1 – FR-27.3, FR-29.1, FR-29.2.

Sign in as **admin** unless a step says otherwise. Alerts are evaluated hourly by the worker; **Check now** on the alerts page does it immediately, which is what these steps use.

### 1. The bell and the centre (FR-27.3)

---

- [ ] A bell sits in the header, next to the search
- [ ] With nothing outstanding it shows no number
- [ ] **Alerts** appears in the sidebar under Dashboard
- [ ] The page opens with two tabs: the alerts, and what you are told about

### 2. Each of the seven triggers (FR-27.1)

---

Make each condition, then press **Check now**.

- [ ] **Low stock** — set a product's minimum above what is on the shelf. The alert names the product, the branch and the minimum
- [ ] A product at **zero** is marked more urgently than one merely low
- [ ] **Customer overdue** — a credit bill past its due date. The alert names the customer, the amount and how many days
- [ ] **Supplier due** — money owed for longer than the threshold. *No branch on this one: owing a supplier is the business's problem*
- [ ] **Cash mismatch** — close a day with the counted cash deliberately wrong. The alert is marked critical

- [ ] **Day not closed** — a branch that took money on a past day with no closing. *Today never appears: it is still open by definition*
- [ ] **Stock adjusted** — make an adjustment; the alert carries the reason and the note
- [ ] **Warranty expiring** — a handset in stock whose cover ends within 30 days

### 3. It does not shout (FR-27.1)

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- [ ] Press **Check now** twice more — no duplicates appear
- [ ] Fix one condition (restock the product), then **Check now** — the alert disappears
- [ ] Break it again and **Check now** — it comes back. *An alert that could only ever fire once would be worse than none*

### 4. Who sees what (FR-27.2)

---

- [ ] As **admin**, alerts are grouped by branch
- [ ] As a **branch manager**, only their branch appears — and no heading, because there is only one
- [ ] As **staff**, low stock and adjustments appear; supplier dues, customer dues and cash mismatches do not. *An alert nobody may act on is noise and a leak*
- [ ] On the settings tab, staff see only the kinds they can act on
- [ ] Staff cannot turn a rule off for the shop — only mute it for themselves

### 5. Reading is personal

---

- [ ] Mark one alert read — the bell's count drops
- [ ] **Mark all read** — the count goes
- [ ] Sign in as another user: **their** count is unchanged. *One manager clearing the bell must not hide a till shortage from the owner*
- [ ] **Everything, including dealt with** shows what was read and resolved

### 6. What you are told about

---

- [ ] **Mute** a kind — the button changes and the shop's setting is untouched
- [ ] As admin, turn a kind **off for the shop** — its existing alerts disappear
- [ ] Change a threshold (say overdue from 7 days to 30) and **Check now** — fewer alerts

### 7. Warranty (FR-29)

---

- [ ] A purchase line asks for **Warranty (months)** and **Warranty by**
- [ ] Book in a handset with 12 months from a named provider
- [ ] Its device page shows the expiry, the length, the provider and whether it is **still in warranty** — without you doing the arithmetic
- [ ] **Inventory** → **Warranty** lists handsets by how soon cover ends

- [ ] An unsold handset shows **In stock** as its owner
- [ ] Sell it, refresh, and the same row now names the **customer**, linking to them. *"Still covered" is only useful beside "whose is it?"*
- [ ] A handset imported from a file with **Warranty** and **Warranty By** columns carries both
- [ ] The window (7 / 30 / 60 / 90 days / a year) is in the URL and can be shared
- [ ] One already expired is marked as such
- [ ] A branch user sees only their branch's handsets

## What M13 does not do

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- **No email or WhatsApp** (PRD OQ-6). In-app only in v1, as FR-27.3 asks. A channel that leaves the building is a decision about cost, a provider and consent — every alert is already a row a channel could read
- **No per-alert snooze**; mute is per kind, for you
- **No job queue**. The worker is an interval loop: both jobs are idempotent and safe to miss and repeat. The first job that must not be lost is when that changes

## M14 — Backup, Data Protection, Hardening & Go-Live

**Delivers.** PRD FR-31.2, FR-31.3, FR-32.1 – FR-32.3, and §9 non-functional requirements.

Most of this module is checked by scripts and tests rather than by clicking. What is left for a person is the part no test can do: proving the restore works on the real server, and watching the shop use it.

### 1. Backups (FR-32.1)

---

On the **server**, not a developer's machine:

- [ ] `./scripts/backup.sh` completes and prints a size and a table count
- [ ] The dump lands in R2 (`restic snapshots` lists it)
- [ ] Break it on purpose — stop the database and run it again. It **fails loudly**, non-zero. *A backup script that exits 0 after a failed dump is worse than none*
- [ ] The cron line is in place and has run unattended at least once
- [ ] A failed run reaches you (email or Sentry), not just a log file

### 2. The restore drill (FR-32.1)

---

- [ ] `./scripts/restore-drill.sh` passes
- [ ] It reports plausible counts — sales, devices, ledger, cash
- [ ] It reports **append-only guards enforce: true**
- [ ] **Write down how long it took**. That is how long the shop is shut for
- [ ] Do it once more from an R2 snapshot, not a local file. *The local copy is not the backup that survives the server*

- [ ] Agree the acceptable data-loss window with the shop: four-hourly dumps lose up to four hours. WAL archiving takes it to about five minutes

### 3. The owner's export (FR-32.3)

---

- [ ] **Settings** → **Your data** shows the row count and what it contains
- [ ] **Download everything** produces a CSV with a section per table
- [ ] It opens in Excel with accents intact
- [ ] Customers, devices, sales and payments are all in it, as data
- [ ] It contains **no password hashes** and no reset tokens
- [ ] It says plainly that it is your data, not a system backup
- [ ] The export appears in **Recent exports** afterwards
- [ ] Signed in as a manager (not the owner), the page redirects away

### 4. Nothing is destroyed (FR-31.2, FR-31.3)

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- [ ] Void a sale — it stays, marked voided, with its reason
- [ ] Reverse a purchase — the handsets become voided, and their **history is still there**
- [ ] Look up one of those IMEIs — the device page still tells its whole story
- [ ] Book the same IMEI in again on a corrected purchase. *Only the claim on the number was released, not the history*
- [ ] Void an expense and a customer payment — both stay, marked
- [ ] There is no button anywhere that deletes money or stock

### 5. Hardening

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- [ ] Get the password wrong twenty times quickly — it starts refusing with a wait, before the account lock
- [ ] **Sign in correctly as many times as you like** — a successful sign-in is never counted. *All the shop's staff share one connection; counting good logins would lock the shop out of its own till after a power cut*
- [ ] Try from a different device — **that** person can still sign in. *One shop being attacked must not lock out another*
- [ ] Wait out the window and sign in normally
- [ ] Request a password reset repeatedly — it is limited the same way
- [ ] A file link (</api/files/...>) opened while signed out is refused
- [ ] The same link, copied and used an hour later, is refused
- [ ] `npm audit` shows no high or critical advisories

### 6. Speed on real data (§9.1)

---

Against a full-size database (`scripts/perf-seed.sql`, or the shop's own after go-live):

- [ ] Global search on a full IMEI — under half a second
- [ ] A partial IMEI — under half a second

- [ ] A device history page — under a second and a half
- [ ] The dashboard — under two seconds
- [ ] Twelve months of analytics — under five seconds
- [ ] Do these at the shop's busiest hour, not at midnight

## 7. Go-live

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- [ ] Everything in the Go-Live Checklist (docs/04 §11)
- [ ] Existing data loaded in the order in docs/04 §10, each step checked
- [ ] Staff trained: billing, returns, dues, the daily closing
- [ ] **Two weeks running in parallel with paper**, one branch
- [ ] At the end of it, cash, stock and dues agree with the paper — or every difference is explained. *This is the real acceptance test*

## What M14 does not do

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- **No WAL archiving yet.** Four-hourly dumps mean losing at most four hours. Point-in-time recovery (about five minutes) is roughly half a day of [pgBackRest](#) setup on the server — recommended for money data, and the decision is the shop's
- **Rate limiting is per process.** One app container, so this is the whole mechanism; scaling to a second container means moving it to Redis
- **No automated penetration testing.** The endpoint sweep proves every route declares who may call it; it is not a substitute for someone trying